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Electronic device including bent portion

Abstract

A first board body includes a first board body upper main surface and a first board body lower main surface that are arranged in a vertical direction. A second board body includes a second board body upper main surface and a second board body lower main surface that are arranged in the vertical direction. The second board body is above the first board body such that the first board body upper main surface and the second board body lower main surface face each other. A semiconductor element is mounted on the first board body lower main surface. An antenna is provided on the second board body. A thickness of a metal plate is larger than a thickness of a first conductor pattern. The metal plate is surface-joined to the first board body upper main surface to overlap the semiconductor element.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-175085 filed on Oct. 19, 2020 and is a Continuation Application of PCT Application No. PCT/JP2021/036967 filed on Oct. 6, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an electronic device including a semiconductor element and an antenna.

2. Description of the Related Art

(2) For example, the laminated module described in JP-A-2019-29669 is a known conventional invention that relates to an electronic device. The laminated module includes an LCP laminated board, an LTCC laminated board, an antenna, and an IC chip. The LCP laminated board includes an upper portion, a lower portion, and a coupling portion. The upper portion of the LCP laminated board is provided above the lower portion of the LCP laminated board. The coupling portion of the LCP laminated board is connected to the upper portion of the LCP laminated board and the lower portion of the LCP laminated board. The coupling portion of the LCP laminated board is bent in the vertical direction. Accordingly, the LCP laminated board includes a U-shape turned 90 degrees clockwise.

(3) The antenna is provided on the upper main surface of the upper portion of the LCP laminated board. The LTCC laminated board is provided on the upper main surface of the lower portion of the LCP laminated board. The IC chip is provided on the upper main surface of the LTCC laminated board. The LTCC laminated board is a rigid board. Accordingly, deformation of the lower portion of the LCP laminated board is suppressed by the LTCC laminated board. As a result, deviation of the disposition of the IC chip due to the deformation of the lower portion of the LCP laminated board is suppressed. That is, the reliability of mounting the IC chip onto the LCP laminated board is improved.

SUMMARY OF THE INVENTION

(4) It should be noted that, in the laminated module described in JP-A-2019-29669, the distance between the IC chip and the antenna is small. Accordingly, the IC chip may be affected by an RF signal (electromagnetic wave) from the antenna. In addition, noise generated in the IC chip may be emitted through the antenna. Specifically, noise is emitted from the IC to the antenna. The antenna emits this noise.

(5) Accordingly, preferred embodiments of the present invention provide electronic devices that each improve the reliability of mounting a semiconductor element onto a first board and prevent the semiconductor element from being affected by an RF signal from the antenna or prevent noise generated in the semiconductor element from being emitted through the antenna.

(6) An electronic device according to a preferred embodiment of the present invention includes a first board including a first board body and a first conductor pattern; a second board including a second board body, a semiconductor element, an antenna, and a metal plate, in which the first board body includes a first board body upper main surface and a first board body lower main surface that are arranged in a vertical direction, the second board body includes a second board body upper main surface and a second board body lower main surface that are arranged in the vertical direction, the second board body is above the first board body such that the first board body upper main surface and the second board body lower main surface face each other, the semiconductor element is mounted on the first board body lower main surface, the antenna is provided on the second board body, and a thickness of the metal plate is larger than a thickness of the first conductor pattern, and the metal plate is surface-joined to the first board body upper main surface to overlap the semiconductor element in the vertical direction.

(7) In an electronic device according to a preferred embodiment of the present invention, it is

possible to improve the reliability of mounting the semiconductor element onto the first board and prevent the semiconductor element from being affected by the RF signal (electromagnetic wave) from the antenna or prevent noise generated in the semiconductor element from being emitted through the antenna.

(8) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a rear view of an electronic device **10**.
- (2) FIG. 2 is a bottom view of a first board **12**.
- (3) FIG. 3 is a top view of a second board **14**.
- (4) FIG. 4 is a rear view of an electronic device **10a**.
- (5) FIG. 5 is a rear view of an electronic device **10b**.
- (6) FIG. 6 is a rear view of an electronic device **10c**.
- (7) FIG. 7 is a rear view of an electronic device **10d**.
- (8) FIG. 8 is a rear view of an electronic device **10e**.
- (9) FIG. 9 is a rear view of an electronic device **10f**.
- (10) FIG. 10 is a rear view of an electronic device **10g**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Preferred Embodiments

(11) Structure of Electronic Device

(12) The structure of an electronic device **10** according to a preferred embodiment of the present invention will be described below with reference to the drawings. FIG. 1 is a rear view of the electronic device **10**. FIG. 2 is a bottom view of a first board **12**. FIG. 3 is a top view of a second board **14**.

(13) In this specification, directions are defined as described below. A direction normal to a first board body upper main surface **S1** is defined as the vertical direction. In addition, two directions orthogonal to the vertical direction are defined as the left-right direction and the longitudinal direction. The left-right direction is orthogonal to the longitudinal direction.

(14) In the following description, X represents a component or a member of the electronic device **10**. Unless otherwise specified, portions of X are defined as described below in this specification. The front portion of X represents the front half of X. The rear portion of X represents the rear half of X. The left portion of X represents the left half of X. The right portion of X represents the right half of X. The upper portion of X represents the upper half of X. The lower portion of X represents the lower half of X. The front end of X represents the end in the forward direction of X. The rear end of X represents the end in the backward direction of X. The left end of X represents the end in the left direction of X. The right end of X represents the end in the right direction of X. The upper end of X represents the end in the upward direction of X. The lower end of X represents the end in the downward direction of X. The front-end portion of X represents the front end of X and the vicinity thereof. The rear-end portion of X represents the rear end of X and the vicinity thereof. The left end portion of X represents the left end of X and the vicinity thereof. The right end portion of X represents the right end of X and the vicinity thereof. The upper end portion of X represents the upper end of X and the vicinity thereof. The lower end portion of X represents the lower end of X and the vicinity thereof.

(15) First, the structure of the electronic device **10** will be described with reference to FIGS. 1 and 3. The electronic device **10** is, for example, a mobile communication terminal such as a

smartphone. The electronic device **10** includes a first board **12**, a second board **14**, antennas **22a** to **22c**, a coupling portion **24**, a semiconductor element **26**, a plurality of electronic components **28**, a metal plate **32**, an insulating resin layer **34**, a plurality of connection conductors **36**, a spacer **38**, and a housing **40**.

(16) The first board **12**, the second board **14**, and the coupling portion **24** are included in one resin board **50**. The resin board **50** includes one or more insulator layers across the first board **12**, the second board **14**, and the coupling portion **24**. The first board **12** is a resin multi-layer board. Specifically, as illustrated in FIG. **1**, the first board **12** includes a first board body **16** and a plurality of first conductor patterns **18** and **19**. The first board body **16** includes a structure in which a plurality of insulator layers are laminated together in the vertical direction. The material of the plurality of insulator layers is, for example, a thermoplastic resin, such as a liquid crystal polymer or a polyimide. As illustrated in FIG. **2**, the first board body **16** has a rectangular or substantially rectangular shape as viewed in the vertical direction, for example. The long sides of the first board body **16** extend in the longitudinal direction. The short sides of the first board body **16** extend in the left-right direction. The first board body **16** includes a first board body upper main surface **S1** and a first board body lower main surface **S2** arranged in the vertical direction. The first board body upper main surface **S1** is located above the first board body lower main surface **S2**. In this specification, “the first board body upper main surface **S1** is located above the first board body lower main surface **S2**” indicates the following state. At least a portion of the first board body upper main surface **S1** is disposed within a region through which the first board body lower main surface **S2** passes when the first board body lower main surface **S2** moves upward in parallel. Accordingly, the first board body upper main surface **S1** may fall in the region through which the first board body lower main surface **S2** passes when the first board body lower main surface **S2** moves upward in parallel, or the first board body upper main surface **S1** may project from the region through which the first board body lower main surface **S2** passes when the first board body lower main surface **S2** moves upward in parallel. In the present preferred embodiment, the first board body upper main surface **S1** fall in the region through which the first board body lower main surface **S2** passes when the first board body lower main surface **S2** moves upward in parallel.

(17) The plurality of first conductor patterns **18** are provided on the first board body lower main surface **S2**. Each of the plurality of first conductor patterns **18** preferably has a rectangular or substantially rectangular shape as viewed in the vertical direction. The plurality of first conductor patterns **18** are mounting electrodes for mounting a semiconductor element **26**, which will be described later.

(18) The plurality of first conductor patterns **19** are provided on the first board body upper main surface **S1**. Each of the plurality of first conductor patterns **19** preferably has a rectangular or substantially rectangular shape as viewed in the vertical direction. The plurality of first conductor patterns **19** are electrically connected to the metal plate **32**, which will be described later. The plurality of first conductor patterns **18** and **19** are formed by, for example, patterning a copper foil by a photolithography process or the like.

(19) The second board **14** is a resin multi-layer board. Specifically, the second board **14** includes a second board body **20** as illustrated in FIG. **1**. The second board body **20** includes a structure in which a plurality of insulator layers are laminated together in the vertical direction. The material of the plurality of insulator layers is, for example, a thermoplastic resin, such as a liquid crystal polymer or a polyimide. As illustrated in FIG. **3**, the second board body **20** preferably has a rectangular or substantially rectangular shape as viewed in the vertical direction. The long sides of the second board body **20** extend in the longitudinal direction. The short sides of the second board body **20** extend in the left-right direction. The thickness of the second board **14** is smaller than the thickness of the first board **12**. The second board body **20** includes a second board body upper main surface **S11** and a second board body lower main surface **S12** that are arranged in the vertical direction. The second board body upper main surface **S11** is located above the second board body

lower main surface **S12**. The second board body **20** is disposed above the first board body **16** such that the first board body upper main surface **S1** and the second board body lower main surface **S12** face each other. In this specification, “the first board body upper main surface **S1** and the second board body lower main surface **S12** face each other” indicates, for example, the following two conditions. The first condition is that the angle defined by the normal vector of the first board body upper main surface **S1** and the normal vector of the second board body lower main surface **S12** is about 180° , for example. However, the directions of these two normal vectors may deviate from about 180° within a range of about $\pm 45^\circ$, for example. The second condition is that the second board body **20** overlaps the first board body **16** in the direction (that is, in the vertical direction) of the normal vector of the first board body upper main surface **S1**.

(20) As illustrated in FIG. 1, the coupling portion **24** is bent in the vertical direction to couple the first board body **16** and the second board body **20** to each other. The coupling portion **24** is curved to project in the right direction as viewed in the longitudinal direction. The coupling portion **24** is more likely than the first board body **16** and the second board body **20** to bend in the vertical direction. Accordingly, the thickness of the coupling portion **24** is smaller than the thickness of the first board body **16** and the thickness of the second board body **20**.

(21) The semiconductor element **26** is an RFIC (radio frequency integrated circuit). Accordingly, the semiconductor element **26** generates a transmission signal obtained by modulating a carrier wave having a predetermined frequency by transmission data and outputs the transmission signal to the antennas **22a** to **22c**. The semiconductor element **26** obtains reception data by demodulating the reception signal received by the antennas **22a** to **22c**. The semiconductor element **26** is mounted on the first board body lower main surface **S2**. More specifically, as illustrated in FIG. 1, the semiconductor element **26** is mounted on the first conductor patterns **18** via solder bumps.

(22) As illustrated in FIG. 1, the plurality of electronic components **28** are mounted on the first board body lower main surface **S2** of the first board body **16**. The plurality of electronic components **28** are, for example, electronic chip components and semiconductor elements.

(23) The antennas **22a** to **22c** emit, as an electromagnetic wave, the transmission signal output by the semiconductor element **26**. The antennas **22a** to **22c** output the received electromagnetic wave to the semiconductor element **26** as the reception signal. Accordingly, the antennas **22a** to **22c** are electrically connected to the semiconductor element **26**. The antennas **22a** to **22c** are patch antennas. As illustrated in FIG. 1, the antennas **22a** to **22c** are provided on the second board body **20**. In the present preferred embodiment, the antennas **22a** to **22c** include the second conductor patterns provided on the second board body upper main surface **S11**. As illustrated in FIG. 3, each of the antennas **22a** to **22c** preferably has a square or substantially square shape as viewed in the vertical direction. As illustrated in FIG. 3, the antennas **22a** to **22c** are arranged in a line from back to front in this order. The antennas **22b** and **22c** overlap the semiconductor element **26** in the vertical direction. The antennas **22a** to **22c** are formed by, for example, patterning a copper foil by a photolithography process or the like.

(24) As illustrated in FIG. 2, the metal plate **32** is a plate-shaped member having a rectangular or substantially rectangular shape as viewed in the vertical direction. A Young's modulus of the material of the metal plate **32** is higher than a Young's modulus of the material of the first board body **16**. The material of the metal plate **32** is, for example, SUS (steel use stainless), a copper alloy (spring material), or the like. As illustrated in FIG. 1, the thickness of the metal plate **32** is larger than the thickness of the first conductor patterns **18** and **19**. Accordingly, the metal plate **32** is less likely than the first board **12** to deform. The metal plate **32** is surface-joined to the first board body upper main surface **S1** so as to overlap the semiconductor element **26** in the vertical direction. Specifically, the insulating resin layer **34** is an adhesive layer through which the metal plate **32** is surface-joined to the first board body upper main surface **S1**. In the surface-joint, the region in which the metal plate **32** and the first board body upper main surface **S1** are joined to each other is planar as viewed in the vertical direction. In the surface-joint, the length in the longitudinal

direction of the insulating resin layer **34** and the length in the left-right direction of the insulating resin layer **34** are larger than the thickness in the vertical direction of the insulating resin layer **34**. The insulating resin layer **34** is, for example, an underfill including a thermosetting acrylic or epoxy resin. Accordingly, the insulating resin layer **34** is provided between the metal plate **32** and the first board body upper main surface **S1**. In addition, the metal plate **32** overlaps substantially the entire first board body upper main surface **S1** in the vertical direction. Accordingly, the metal plate **32** overlaps the entire semiconductor element **26** in the vertical direction. In addition, the metal plate **32** overlaps the antennas **22a** to **22c** in the vertical direction. Accordingly, the metal plate **32** is disposed between the antennas **22a** to **22c** and the semiconductor element **26**.

(25) However, as illustrated in FIG. **1**, the metal plate **32** is electrically connected to the plurality of first conductor patterns **19**. More specifically, the plurality of connection conductors **36** are provided on the plurality of first conductor patterns **19**, respectively. The plurality of connection conductors **36** are made of, for example, solder. The metal plate **32** is electrically connected to the plurality of first conductor patterns **19** via the plurality of connection conductors **36**. Accordingly, the metal plate **32** is connected to the ground potential.

(26) As illustrated in FIG. **1**, the spacer **38** is provided between the second board body lower main surface **S12** and the metal plate **32**. As a result, the spacer **38** maintains a distance between the second board body lower main surface **S12** and the metal plate **32**. The spacer **38** is a plate-shaped member. The upper main surface of the spacer **38** is in contact with the second board body lower main surface **S12**. The lower main surface of the spacer **38** is in contact with the upper main surface of the metal plate **32**. The spacer **38** may be a member that includes only a function (spacer function) of maintaining the distance between the second board body lower main surface **S12** and the metal plate **32** or may be a member that includes an additional function other than the spacer function. The additional function is, for example, a display panel or a battery.

(27) As illustrated in FIG. **1**, the housing **40** accommodates the first board **12**, the second board **14**, the antennas **22a** to **22c**, the coupling portion **24**, the semiconductor element **26**, the plurality of electronic components **28**, the metal plate **32**, the insulating resin layer **34**, the plurality of connection conductors **36**, and the spacer **38**. The housing **40** is the display panel, the case, or the like of the electronic device **10**. When the housing **40** is the display panel, the housing **40** is a glass plate. When the housing **40** is the case, the housing **40** is a metal plate, a glass plate, or a resin plate.

(28) Effect

(29) In the electronic device **10**, the reliability of mounting the semiconductor element **26** onto the first board **12** is improved. More specifically, when the first board body **16** deforms, stress is applied to the connection portion between the first board body **16** and the semiconductor element **26**. Such stress causes the connection portion between the first board **12** and the semiconductor element **26** to break. That is, such stress causes the electrical disconnection between the first board **12** and the semiconductor element **26** to break. Accordingly, in the electronic device **10**, the metal plate **32** is surface-joined to the first board body upper main surface **S1**. This causes the metal plate **32** to reinforce the first board body **16**. Accordingly, the metal plate **32** reduces or prevents the deformation of the first board body **16**. As a result, application of stress to the connecting portion between the first board body **16** and the semiconductor element **26** is reduced or prevented. Therefore, in the electronic device **10**, the reliability of mounting the semiconductor element **26** onto the first board **12** is improved.

(30) In the electronic device **10**, it is possible to prevent the semiconductor element **26** from being affected by the RF signal (electromagnetic wave) from the antennas **22a** to **22c** or prevent noise generated in the semiconductor element **26** from being emitted through the antennas **22a** to **22c**. More specifically, the second board body **20** is disposed above the first board body **16** such that the first board body upper main surface **S1** and the second board body lower main surface **S12** face each other. The semiconductor element **26** is mounted on the first board body lower main surface

S2. The antennas **22a** to **22c** are provided on the second board body **20**. In addition, the metal plate **32** is surface-joined to the first board body upper main surface **S1** so as to overlap the semiconductor element **26** in the vertical direction. Accordingly, the metal plate **32** is located between the antennas **22a** to **22c** and the semiconductor element **26**. Therefore, the RF signal (electromagnetic wave) emitted through the antennas **22a** to **22c** is blocked by the metal plate **32** and less likely to reach the semiconductor element **26**. Similarly, noise emitted by the semiconductor element **26** is blocked by the metal plate **32** and less likely to reach the antennas **22a** to **22c**. Accordingly, in the electronic device **10**, it is possible to prevent the semiconductor element **26** from being affected by the RF signal (electromagnetic wave) from the antennas **22a** to **22c** or reduce or prevent noise generated in the semiconductor element **26** from being emitted through the antennas **22a** to **22c**. In particular, the metal plate **32** is connected to the ground potential.

Accordingly, in the electronic device **10**, it is possible to more effectively prevent the semiconductor element **26** from being affected by the RF signal (electromagnetic wave) from the antennas **22a** to **22c** or to more effectively reduce or prevent noise generated in the semiconductor element **26** from being emitted through the antennas **22a** to **22c**.

(31) In the electronic device **10**, the metal plate **32** overlaps the antennas **22a** to **22c** in the vertical direction. Accordingly, in the electronic device **10**, it is possible to more effectively prevent the semiconductor element **26** from being affected by the RF signal (electromagnetic wave) from the antennas **22a** to **22c** or to more effectively reduce or prevent noise generated in the semiconductor element **26** from being emitted through the antennas **22a** to **22c**.

(32) In the electronic device **10**, variations in the impedance in consideration of the antennas **22a** to **22c** and variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented. More specifically, the spacer **38** is provided between the second board body lower main surface **S12** and the metal plate **32** to reduce or prevent changes in the distance between the second board body lower main surface **S12** and the metal plate **32**. This reduces or prevents fluctuations in the distance between the antennas **22a** to **22c** and the metal plate **32**. Accordingly, fluctuations in the capacitance generated between the antennas **22a** to **22c** and the metal plate **32** are reduced or prevented. Accordingly, in the electronic device **10**, variations in the impedance in consideration of the antennas **22a** to **22c** and variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented.

(33) In the electronic device **10**, variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented. More specifically, the coupling portion **24** is more likely than the first board body **16** and the second board body **20** to bend in the vertical direction. To achieve this, in the electronic device **10**, the thickness of the coupling portion **24** is smaller than the thickness of the first board body **16** and the thickness of the second board body **20**. Accordingly, the coupling portion **24** is more likely to deform. Furthermore, the stress applied by the coupling portion **24** to the first board body **16** and the second board body **20** when the coupling portion **24** deforms is reduced. Accordingly, the second board body **20** is prevented from being deformed by the stress applied by the coupling portion **24** to the second board body **20** when the coupling portion **24** deforms. The deformation of the second board body **20** reduces or prevents the distance between the antennas **22a** to **22c** and the metal plate **32** from changing. As a result, in the electronic device **10**, variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented. In addition, the breakage of the resin board **50** is reduced or prevented.

(34) In the electronic device **10**, the heat generated in the first board **12** and the semiconductor element **26** diffuses via the metal plate **32**. This improves the cooling efficiency of the first board **12** and the semiconductor element **26**.

(35) First Modification

(36) An electronic device **10a** according to a first modification will be described below with reference to the drawings. FIG. **4** is a rear view of the electronic device **10a**.

(37) The electronic device **10a** differs from the electronic device **10** in that the electronic device

10a includes a support member **60** instead of the spacer **38**. More specifically, the housing **40** supports the second board **14**. The support member **60** is supported by the housing **40**. The support member **60** supports the first board **12**. In addition, the support member **60** overlaps the first board **12** and does not overlap the second board **14** in the vertical direction. The other structure of the electronic device **10a** is the same as that of the electronic device **10**, and accordingly, description thereof is omitted.

(38) In the electronic device **10a**, variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented and losses due to the spacer **38** are reduced or prevented. More specifically, the housing **40** supports the second board **14**. The support member **60** is supported by the housing **40**. The support member **60** supports the first board **12**. Accordingly, the support member **60** reduces or prevents changes in the distance between the second board body lower main surface **S12** and the metal plate **32**. Accordingly, fluctuations in the distance between the antennas **22a** to **22c** and the metal plate **32** are reduced or prevented. Accordingly, fluctuations in the capacitance generated between the antennas **22a** to **22c** and the metal plate **32** are reduced or prevented. In the electronic device **10a**, variations in the emission characteristics of the antennas **22a** to **22c** are reduced or prevented. Furthermore, the support member **60** overlaps the first board **12** and does not overlap the second board **14** in the vertical direction. Accordingly, the support member **60** is not present between the antennas **22a** to **22c** and the metal plate **32**. That is, there is an air layer between the antennas **22a** to **22c** and the metal plate **32**. An increase in the dielectric constant in the region between the antennas **22a** to **22c** and the metal plate **32** is reduced or prevented. As a result, in the electronic device **10a**, losses due to the spacer **38** are reduced. At this time, the metal plate **32** can be used as the ground for the antennas **22a** to **22c**.

(39) Second Modification

(40) An electronic device **10b** according to a second modification will be described below with reference to the drawings. FIG. 5 is a rear view of the electronic device **10b**.

(41) The electronic device **10b** differs from the electronic device **10a** in that the electronic device **10b** further includes a frame **70** and a thermal conductor **72**. The frame **70** is the internal skeleton of the electronic device **10b**. Accordingly, the frame **70** supports components (not illustrated) provided within the housing **40**. However, the frame **70** may support the resin board **50**. The material of the frame **70** is, for example, a metal such as aluminum.

(42) The thermal conductor **72** is a film through which the metal plate **32** and the frame **70** are connected to each other. The thermal conductor **72** is attached to the metal plate **32** and the frame **70** with an adhesive or the like. The material of the thermal conductor **72** includes high thermal conductivity. The material of the thermal conductor **72** is, for example, graphite, copper, or aluminum. Accordingly, the heat generated in the first board **12** and the semiconductor element **26** is transferred to the frame **70** via the metal plate **32** and the thermal conductor **72**. This improves the cooling efficiency of the first board **12** and the semiconductor element **26** more effectively.

(43) In addition, when the thermal conductor **72** is made of a conductive material, the frame **70** (metal component) is electrically connected to the metal plate **32**. The frame **70** is connected to the ground potential. Accordingly, the potential of the metal plate **32** becomes more stable. It should be noted that the other structure of the electronic device **10b** is the same as that of the electronic device **10a**, and accordingly, description thereof is omitted.

(44) Third Modification

(45) An electronic device **10c** according to a third modification will be described below with reference to the drawings. FIG. 6 is a rear view of the electronic device **10c**.

(46) The electronic device **10c** differs from the electronic device **10** in that the electronic device **10c** includes an anisotropic conductive film **80** instead of the insulating resin layer **34**. More specifically, the anisotropic conductive film **80** is commonly referred to as an ACF (anisotropic conductive film). The anisotropic conductive film **80** is a conductive and insulating film-type joint material having a structure in which conductive particles are uniformly dispersed in a thermosetting

resin. The anisotropic conductive film **80** as described above can connect a plurality of terminals and a plurality of terminals to each other at the same time. The anisotropic conductive film **80** surface-joins the metal plate **32** to the first board body upper main surface **S1** and electrically connects the metal plate **32** and the first conductor patterns **19** to each other. The first conductor patterns **19** project upward from the first board body upper main surface **S1**. Accordingly, the pressure applied to the portion of the anisotropic conductive film **80** located between the first conductor patterns **19** and the metal plate **32** is larger than the pressure applied to the portion other than the portion of the anisotropic conductive film **80** located between the first conductor patterns **19** and the metal plate **32**. Accordingly, the first conductor patterns **19** and the metal plate **32** are electrically connected to each other via the conductive particles included in the anisotropic conductive film **80**. It should be noted that the other structure of the electronic device **10c** is the same as that of the electronic device **10**, and accordingly, description thereof is omitted.

(47) Fourth Modification

(48) An electronic device **10d** according to a fourth modification will be described below with reference to the drawings. FIG. **7** is a rear view of the electronic device **10d**.

(49) The electronic device **10d** differs from the electronic device **10** in that the electronic device **10d** includes connectors **90** and **92** instead of the coupling portion **24**. The connector **90** is mounted on the first board body upper main surface **S1** of the first board body **16**. The connector **92** is mounted on the second board body lower main surface **S12** of the second board body **20**. The connector **90** and the connector **92** are electrically connected to each other. Accordingly, the semiconductor element **26** and the antennas **22a** to **22c** are electrically connected to each other via the first board **12**, the connectors **90** and **92**, and the second board **14**. The other structure of the electronic device **10d** is the same as that of the electronic device **10**, and accordingly, description thereof is omitted.

(50) Fifth Modification

(51) An electronic device **10e** according to a fifth modification will be described below with reference to the drawings. FIG. **8** is a rear view of the electronic device **10e**.

(52) The electronic device **10e** differs from the electronic device **10** in that the second board body **20** and the coupling portion **24** are electrically connected to each other via the connectors **90** and **92**. That is, the connector **90** is mounted on the outer peripheral surface of the coupling portion **24**. The connector **92** is mounted on the second board body lower main surface **S12**. The connector **90** is electrically connected to the connector **92**. The other structure of the electronic device **10e** is the same as that of the electronic device **10**, and accordingly, description thereof is omitted. The electronic device **10e** can achieve the same effect as the electronic device **10**.

(53) In addition, since the electronic device **10e** can be disposed three-dimensionally, the storage space of the electronic device **10e** can be reduced. In addition, since the coupling portion **24** can be connected to the thin plate-shaped second board body **20**, the gap between the second board body **20** and the housing **40** can be reduced. In addition, since the coupling portion **24** can be detached from the second board body **20**, components can be easily replaced.

(54) Since the thickness of the coupling portion **24** is small in the electronic device **10e**, the curvature radius of the coupling portion **24** can be reduced. Accordingly, the coupling portion **24** can be disposed in a small space. In addition, since the thickness of the coupling portion **24** is small, the coupling portion **24** is easily bent. Accordingly, the force exerted by the coupling portion **24** on the connector **90** is small. As a result, an unnecessary force is less likely to be applied to the portion between the connector **90** and the connector **92**.

(55) Sixth Modification

(56) An electronic device **10f** according to a sixth modification will be described below with reference to the drawings. FIG. **9** is a rear view of the electronic device **10f**.

(57) The electronic device **10f** differs from the electronic device **10e** in that the thickness of the first board body **16**, the thickness of the second board body **20**, and the thickness of the coupling portion

24 are identical to each other. The other structure of the electronic device **10f** is the same as that of the electronic device **10e**, and accordingly, description thereof is omitted.

(58) In addition, since the electronic device **10f** can be disposed three-dimensionally, the storage space of the electronic device **10f** can be reduced. In addition, since the coupling portion **24** can be connected to the thin plate-shaped second board body **20**, the gap between the second board body **20** and the housing **40** can be reduced. In addition, since the coupling portion **24** can be detached from the second board body **20**, components can be easily replaced.

(59) Seventh Modification

(60) An electronic device **10g** according to a seventh modification will be described below with reference to the drawings. FIG. **10** is a rear view of the electronic device **10g**.

(61) The electronic device **10g** differs from the electronic device **10** in that the thickness of the first board body **16**, the thickness of the second board body **20**, and the thickness of the coupling portion **24** are identical to each other and the first board body **16** and the second board body **20** are bent in the vertical direction. More specifically, the thickness of the first board body **16**, the thickness of the second board body **20**, and the thickness of the coupling portion **24** are identical to each other. This causes the first board body **16** and the second board body **20** to easily bend in the vertical direction similarly to the coupling portion **24**. In addition, there is no step at the boundary between the first board body **16** and the coupling portion **24**. There is no step at the boundary between the second board body **20** and the coupling portion **24**. This prevents the resin board **50** from being broken at the boundary between the first board body **16** and the coupling portion **24** and at the boundary between the second board body **20** and the coupling portion **24**. In addition, since there is no need to form a cavity in the resin board **50**, the structure of the resin board **50** becomes simple. As a result, the resin board **50** can be manufactured easily. However, the metal plate **32** is provided on the first board body upper main surface **S1**. Accordingly, a first board body **16** portion in which the metal plate **32** is not provided is bent in the vertical direction. It should be noted that the coupling portion **24** may have a structure in which the coupling portion **24** is easily bent in the vertical direction by, for example, reducing the number of conductor layers of the coupling portion **24**.

(62) When the thickness of the first board body **16** and the thickness of the second board body **20** are identical to the thickness of the coupling portion **24** as described above, the first board body **16** and the second board body **20** become thinner. As a result, the resin board **50** can be disposed in a narrow space by bending the resin board **50**.

Other Preferred Embodiments

(63) The electronic devices according to preferred embodiments of the present invention are not limited to the electronic devices **10** and **10a** to **10g** and can be modified within the scope of the concept. It should be noted that the structures of the electronic devices **10** and **10a** to **10g** may be combined arbitrarily.

(64) The antennas **22a** to **22c** need only be provided in the second board body **20**, and may be provided inside the second board body **20** or on the second board body lower main surface **S12**.

(65) The antennas **22a** to **22c** need not be formed of the conductor patterns of the second board **14**. The antennas **22a** to **22c** may be components mounted on the second board body **20**.

(66) Each of the electronic devices **10** and **10a** to **10g** includes three antennas, for example. However, each of the electronic devices **10** and **10a** to **10g** may have one or two antennas or four or more antennas.

(67) The metal plate **32** need not overlap the antennas **22a** to **22c** in the vertical direction.

(68) The metal plate **32** need not overlap the entire semiconductor element **26** in the vertical direction. That is, the metal plate **32** may overlap a portion of the semiconductor element **26** in the vertical direction. However, the metal plate **32** overlaps the entire semiconductor element **26** in the vertical direction. Accordingly, it is possible to more effectively reduce or prevent the semiconductor element **26** from being affected by the RF signal (electromagnetic wave) from the

antennas **22a** to **22c** or to more effectively reduce or prevent the antennas **22a** to **22c** from being affected by noise of the semiconductor element **26**.

(69) The antennas **22a** to **22c** emit the transmission signal output by the semiconductor element **26** as the electromagnetic wave and output the received electromagnetic wave to the semiconductor element **26** as the reception signal. However, the antennas **22a** to **22c** may either emit the transmission signal output by the semiconductor element **26** as the electromagnetic wave or output the received electromagnetic wave to the semiconductor element **26** as the reception signal.

(70) The antennas **22a** to **22c** need not be electrically connected to the semiconductor element **26**. In addition, the semiconductor element **26** need not be an RFIC. That is, the antennas **22a** to **22c** need not emit the transmission signal output by the semiconductor element **26** as the electromagnetic wave and need not output the received electromagnetic wave to the semiconductor element **26** as the reception signal.

(71) The first board **12** may include conductor patterns inside the first board body **16**, on the first board body upper main surface **S1**, or on the first board body lower main surface **S2**. The second board **14** may have conductor patterns inside the second board body **20**, on the second board body upper main surface **S11**, or on the second board body lower main surface **S12**. The coupling portion **24** may include conductor patterns inside the coupling portion **24** or on two main surfaces of the coupling portion **24**.

(72) The first board **12** and the second board **14** may be or need not be flexible.

(73) In the electronic device **10**, the first conductor patterns **18** and the metal plate **32** may be electrically connected to each other by surface-joining the first board body upper main surface **S1** and the metal plate **32** to each other by using conductive joint materials instead of the insulating resin layer **34** and the connection conductor **36**.

(74) It should be noted that the housing **40** supports the second board **14** in the electronic device **10a**. In addition, the support member **60** supports the first board **12**. However, the housing **40** may support the first board **12**. In addition, the support member **60** may support the second board **14**. That is, the housing **40** need only support one of the first board **12** and the second board **14**. The support member **60** need only support the other of the first board **12** and the second board **14**.

(75) It should be noted that an anisotropic conductive film, solder, or the like may be used instead of the connectors **90** and **92** in the electronic device **10d**. Since the parasitic components caused by the connectors **90** and **92** are eliminated in this case, the reflection of high-frequency signals is reduced or prevented.

(76) It should be noted that the first board body **16** and/or the second board body **20** may be bent in the vertical direction in the electronic devices **10** and **10a** to **10g**.

(77) It should be noted that the electronic device **10a** may further include the spacer **38**. That is, the electronic device **10a** may have both the spacer **38** and the support member **60**. This makes the distance between the antennas **22a** to **22c** and the metal plate **32** constant. As a result, variations in the capacitance between the antennas **22a** to **22c** and the metal plate **32** connected to the ground potential are reduced. In addition, the antennas **22a** to **22c** can be positioned with respect to the housing **40**. Accordingly, variations in the distance between the antennas **22a** to **22c** and the housing **40** are reduced.

(78) It should be noted that the connector **92** may be mounted on the second board body upper main surface **S11** in the electronic devices **10e** and **10f**.

(79) It should be noted that, in the electronic devices **10e** and **10f**, the second board body **20** and the coupling portion **24** may be connected to each other not by the connectors **90** and **92** but by conductive connection members. In this case, the electronic devices **10e** and **10f** can be made thinner.

(80) It should be noted that the thickness of the coupling portion **24** is smaller than the thickness of the first board body **16** and the thickness of the second board body **20** in the electronic device **10e**. Accordingly, a cavity (dug portion) is formed at a position in the resin board **50** that corresponds to

the coupling portion **24**. In this case, the resin board **50** is bent such that the first board body lower main surface **S2** is continuous with the outer peripheral surface of the coupling portion **24**. This reduces or prevents breakage from occurring at the boundary between the coupling portion **24** and the first board body **16**. However, the resin board **50** may be bent such that the first board body upper main surface **S1** is continuous with the inner peripheral surface of the coupling portion **24**. (81) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An electronic device comprising: a first board including a first board body and a first conductor pattern; a second board including a second board body; a semiconductor element; an antenna; a metal plate; and at least one of a coupling portion and a connecting member; wherein the first board body includes a first board body upper main surface and a first board body lower main surface that are arranged in a vertical direction; the second board body includes a second board body upper main surface and a second board body lower main surface that are arranged in the vertical direction; the second board body is above the first board body such that the first board body upper main surface and the second board body lower main surface face each other; the semiconductor element is mounted on the first board body lower main surface; the antenna is provided on the second board body; a thickness of the metal plate is larger than a thickness of the first conductor pattern, and the metal plate is surface-joined to the first board body upper main surface to overlap the semiconductor element in the vertical direction; and the first board body and the second board body are included in at least one resin board, and at least one of the first board body, the second board body, and the coupling portion is bent in the vertical direction so as to couple the first board body and the second board body to each other.
2. The electronic device according to claim 1, wherein the metal plate overlaps the antenna in the vertical direction.
3. The electronic device according to claim 1, wherein the first conductor pattern is provided on the first board body upper main surface, and the metal plate is electrically connected to the first conductor pattern.
4. The electronic device according to claim 3, further comprising: a connection conductor provided on the first conductor pattern; and an insulating resin layer through which the metal plate is surface-joined to the first board body upper main surface; wherein the metal plate is electrically connected to the first conductor pattern via the connection conductor.
5. The electronic device according to claim 3, further comprising: an anisotropic conductive film through which the metal plate is surface-joined to the first substrate body upper main surface and the metal plate is electrically connected to the first conductor pattern.
6. The electronic device according to claim 1, further comprising: a housing that supports one of the first board and the second board; and a support that supports the other of the first board and the second board and is supported by the housing.
7. The electronic device according to claim 1, further comprising: a spacer provided between the second board body lower main surface and the metal plate.
8. The electronic device according to claim 1, further comprising: a metal component electrically connected to the metal plate.
9. The electronic device according to claim 1, wherein the antenna is electrically connected to the semiconductor element.
10. The electronic device according to claim 1, wherein the semiconductor element is an RFIC.
11. The electronic device according to claim 10, wherein the antenna is operable to emit a

transmission signal output by the semiconductor element as an electromagnetic wave or the antenna is operable to output a received electromagnetic wave to the semiconductor element as a reception signal.

12. The electronic device according to claim 1, wherein the metal plate is connected to a ground potential.

13. The electronic device according to claim 1, wherein the electronic device is a mobile communication terminal.

14. The electronic device according to claim 1, wherein a thickness of the coupling portion is less than a thickness of the first board and the second board.

15. The electronic device according to claim 1, wherein a plurality of electronic components in addition to the semiconductor element are mounted on the first board body lower main surface.

16. The electronic device according to claim 1, wherein a plurality of antennas as provided in addition to the antenna.

17. The electronic device according to claim 16, wherein the plurality of antennas are electrically connected to the semiconductor element.

18. The electronic device according to claim 16, wherein the plurality of antennas are patch antennas.

19. The electronic device according to claim 16, wherein the plurality of antennas are square or substantially square.
